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Subject: NEMA Comments- “AI Action Plan”

The National Electrical Manufacturers Association (NEMA) respectfully submits the following comments in response to the request for information on the *Development of an Artificial Intelligence (AI) Action Plan* (the Plan), as directed by Executive Order 14179, signed January 23, 2025.

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About NEMA

NEMA represents over 300 electrical equipment manufacturers that make safe, reliable, and efficient products and systems. Together, our members contribute 1% of U.S. GDP and directly provide nearly 460,000 American jobs, contributing more than \$250 billion to the U.S. economy. Our members produce goods for the grid, industrial, built environment, and mobility sectors. The electroindustry is a key driver of infrastructure development and future economic growth.¹

¹ Additional information about NEMA may be found at <https://www.nema.org/>.

Comments

The electroindustry has been rapidly evolving to meet the tremendous market demand for electric goods and components, driven by factors such as electrification, digitalized connectivity throughout the grid and built environment, the rise of the data center and quantum computing industries, and the growth of domestic manufacturing. These conditions, combined with the rapid advancement and implementation of sensory technologies and chip-centric management and operational systems, present both production opportunities and supply chain challenges for electrical manufacturers. AI is one of those technologies.

The use of AI across the electroindustry is not novel, having been used in production processes in some form since the 1950s.² As such, the industry has been at the forefront of developing safe and mature data processing models which have led the world in algorithmic production. Current day market factors are enabling the use of AI well beyond the closed environments of production floors and laboratories to the consuming public through the exponential digitalization of data, the development of Internet of Things (IoT) networks, and the steady improvements in machine learning (ML) and deep learning algorithms for generative purposes.

In the most general sense, AI technology applications in manufacturing introduces scale and efficiency and is best applied to two types of scenarios: data analysis and subsequent predictive recommendations and actions; and routine redundant tasks. In that context, NEMA's comments and recommendations on the Plan are provided below.

AI in Manufacturing

Electromanufacturing companies incorporate connected sensors, devices and systems within their operational environments which generate vast amounts of machine data. AI algorithms are used to derive understanding from concept machine data in aspects such as complexity, quality, and labeling. These ML algorithms can then improve upon the data

² National Electrical Manufacturers Association. (n.d.). *AI in the Electroindustry and Medical Imaging Sectors*. Retrieved from https://www.nema.org/docs/default-source/council-documents-library/documents/ai-electroindustry-ebook.pdf?sfvrsn=d89e2244_0

and knowledge gained to subsequently improve operational processes, enabling efficiencies which boost capacity output for products needed to advance the AI revolution.

Furthermore, AI can help optimize workflows, predict maintenance needs, and increase workplace safety. It can also increase the reliability and resiliency of the supply chain by enabling manufacturers to proactively and accurately react to potential and identified disruptions and challenges. All of these are an imperative as the electroindustry leads the charge in revitalizing domestic manufacturing and production to achieve American policy goals.

Enhanced manufacturing efficiency provides numerous societal benefits, including environmental sustainability, increased product quality and safety, economic growth and job satisfaction, and improved supply chain resilience. The Plan needs to recognize these benefits as being essential to a modern economy and society; it is extremely important that routine and constructive collaboration between the private and public sectors occurs to create a supportive ecosystem for AI adoption and use in the advanced manufacturing sector.

AI in Energy

AI in the energy sector is twofold: it is a tool that allows for the optimization of load management throughout the grid; simultaneously, AI is only enabled by a resilient backbone of electrical grid hardware and software components. AI plays a pivotal role in enabling a smart electric grid by enhancing efficiency, reliability, and resilience. But it can only be made smart (intelligent) with the availability and proper implementation of a combination of new-age and legacy components and products.

AI provides for a whole suite of cutting-edge possibilities to enhance the grid, including but not limited to real-time monitoring and automation systems (predictive maintenance, real-time response), demand response and predictive analytics (distributive energy demand management; load forecasting) energy storage optimization, and cybersecurity. These features empower grid owners and operators to dramatically enhance resilience and ensure load reliability.

Smart grid technologies and sensors can capture information such as the amount of power consumed, the duration of peak use, customer location, and other use signatures which can determine where on the grid power is being drawn. AI can identify specific patterns of energy use by location, customer type, or other criteria. These determinations lead to generation use predictions which can be fed to power production systems, allowing energy producers to balance electricity supply and demand.

The Plan should recognize the importance of AI's role of energy management through innovation and modernization of the electric grid. Enacting manufacturing tax friendly policies can help to spur this.

The ability of AI to scale and provide immense benefits are directly dependent on an electrical component supply chain to facilitate and consistently deliver the overwhelming power needed for the technology to function. This is especially true for data centers, which are integral to AI development. Ample and sustained investment in the data center supply chain is needed to ensure that the demand for vital components, including transformers, switchgears, circuit breakers, panel boards, wires, substations, and other elements, can be met with a consistent and predictable supply.

Additionally, supply chain capacity investment must be complemented by sound public policies and reasonable regulatory processes to maintain such investment and further global leadership. As mentioned above, the electroindustry has pioneered the use of AI in American manufacturing and currently contributes to the United States being at the forefront in AI development and implementation. Federal policies must prioritize America's leadership in the AI space; existing policies should be modernized, and new policies must bolster and maintain this position.

As the supply chain is critical to both data center and AI development, NEMA strongly recommends that the Plan include considerations to ensure market certainty for manufacturers and other supply chain entities who support and promote industries driving AI innovation.

Not all AI is Created Equal

AI is an umbrella term covering numerous computing systems. As referenced above, most AI used in the manufacturing process utilize ML models, which are trained on large data sets with human input, conversations, user queries and responses. These models operate in controlled processing environments, where the data collected and analyzed maintains a high standard of integrity to ensure that statistical outcomes are valid and supported.

The growing popularity of generative AI uses in consumer-facing products should not be confused with ML or other models which are deployed in controlled and monitored settings. Generative AI models are trained on different sets of data to learn predictive patterns to produce various types of content, including text, imagery, audio, and synthetic data.

Governance frameworks for AI need to consider who the end users are and the integrity of data that is being analyzed when being developed. The Plan should understand the known and perceived end-uses of the various AI models so lawmakers and regulators can create appropriate and specific guardrails which further innovation and American leadership while avoiding harmful or illegal outcomes.

Role of AI Standards & Frameworks

AI standards promote the interoperability between different systems and devices to facilitate seamless integration of digital solutions into existing infrastructure, both within and throughout the manufacturing sector, as well as across economic sectors. They are also critical to ensure the reliable and safe interface of connected systems.

The Plan can encourage AI standards development and participation by endorsing and elevating existing internationally recognized standards development organization (SDO) initiatives intended to boost participation in standards-setting activities. This includes award programs, such as those from the American National Standards Institute and the Society for Standardization Professionals, that recognize individuals and organizations for contributions to standards development, which encourage mentorship or career development in standards development, and foster collaboration with small and medium size companies to elevate their voice, needs, and strengths in standards development.

Additionally, the electroindustry recommends that the Plan encourage the voluntary adoption of public, consensus-based, well-respected and -sourced governance, AI risk management frameworks, and other public documents including:

- The NIST AI Risk Management Framework (RMF)³;
- Playbooks stemming from the RMF⁴;
- The use of advanced threat assessment models, including the MITRE Atlas⁵ which can help the industry develop and securely operate resilient and trustworthy AI-enabled systems.
- The Internet of Things (IoT) Advisory Board Report which includes several sections on the growing importance of AI within IoT (AIoT)⁶

Importantly, protecting the intellectual property (IP) and copyrights of SDOs is paramount. The Plan needs to clearly recognize appropriate and legal safeguards and guardrails for existing IP accessed by AI systems.

AI Governance Aspects

AI is a tool to amplify human capabilities, particularly the processing of information to make decisions. In the end, a human ultimately is the recipient of an AI recommendation, and it is a human that is responsible for the outcome of any decision made from that recommendation. In proper governance of AI, to enhance trust and accountability and effectively manage risk, a human in-loop is fundamental.

Additionally, electromanufacturers rely on the integrity of data inputs to trust the recommendations and actions AI systems produce; where data originates from and is sourced matters. In general applications, ML in manufacturing is used in unadulterated

³ <https://www.nist.gov/itl/ai-risk-management-framework>

⁴ <https://airc.nist.gov/airmf-resources/playbook/>

⁵ <https://atlas.mitre.org/>

⁶ [https://www.nist.gov/system/files/documents/2024/10/21/IoT Advisory Board Report](https://www.nist.gov/system/files/documents/2024/10/21/IoT_Advisory_Board_Report)

and non-public environments, where data inputs and outputs are controlled for system integrity. However, as the use of data sets by AI evolves to address new and unforeseen opportunities and challenges in manufacturing, regulatory and legal rules around the collection, handling, and utilization of data need to be clearly defined. The Plan needs to consider these aspects when considering appropriate AI governance models.

Conclusion

NEMA respectfully requests consideration of our comments, and we look forward to working with the NITRD NCO CEC in the development and implementation of the “AI Action Plan.” Should you have any questions or need additional information, please do not hesitate to contact me.

Sincerely,

Spencer Pederson
Senior Vice President, Public Affairs